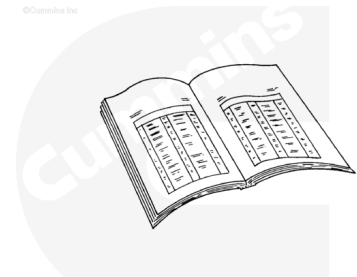


Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.



ck800wa

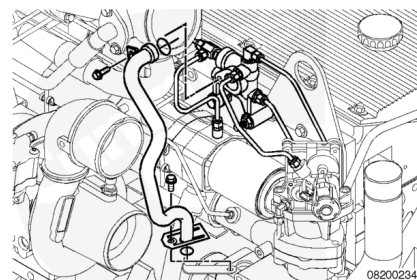
Remove or disconnect the following:

- Drain the cooling system. Refer to Procedure 008-018 (</qs3/pubsys2/xml/en/procedures/35/35-008-018-tr.html>).
- Air inlet pipe to the turbocharger. Refer to Procedure 010-080 (</qs3/pubsys2/xml/en/procedures/35/35-010-080-tr.html>).
- Coolant vent lines. Refer to Procedure 008-017 (</qs3/pubsys2/xml/en/procedures/35/35-008-017-tr.html>).
- EGR valve coolant return line. Refer to Procedure 011-030 (</qs3/pubsys2/xml/en/procedures/35/35-011-030-tr.html>).
- EGR cooler coolant vent lines. Refer to Procedure 011-031 (</qs3/pubsys2/xml/en/procedures/35/35-011-031-tr.html>).
- Turbocharger coolant return line. Refer to Procedure 010-041 (</qs3/pubsys2/xml/en/procedures/35/35-010-041-tr.html>).
- EGR cooler connection heat shield. Refer to Procedure 011-032 (</qs3/pubsys2/xml/en/procedures/35/35-011-032-tr.html>).

Remove

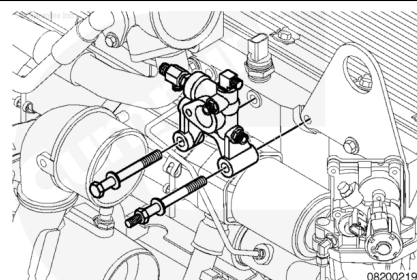
Remove both the water manifold return tube capscrews and water manifold return tube.

Remove and discard both o-rings.



Remove the water manifold mounting capscrew and loose washer as well as studded capscrew and loose washer.

Remove the water manifold.



Disassemble

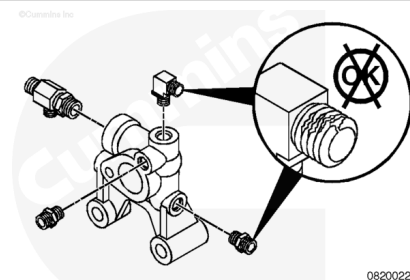
Remove the turbocharger coolant return fitting.

Remove the cylinder head vent fitting.

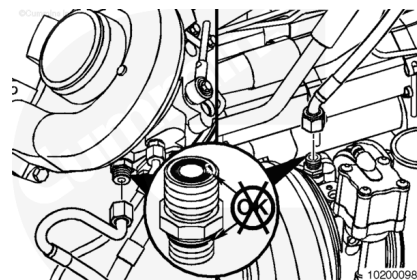
Remove the EGR valve coolant return fitting.

Remove the EGR cooler coolant vent T-fitting.

Inspect threads on all fittings for damage.



Inspect the fittings for damaged o-rings. Replace the o-rings, if damaged.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

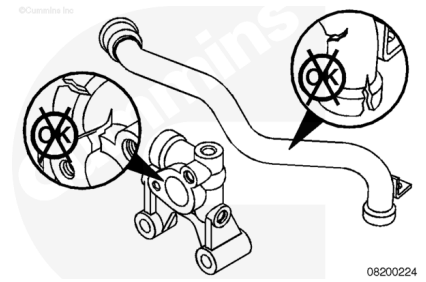
⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Clean the water manifold return tube and water manifold with solvent.

Dry with compressed air.

Inspect the water manifold return tube and water manifold for cracks or damage. Replace the component, if cracked or damaged.



Assemble

⚠ CAUTION ⚠

Do not use petroleum products on the coolant plumbing o-rings. The o-rings will swell and cause the connection to leak.

Lubricate the o-rings with vegetable oil, clean engine coolant, or soapy water.

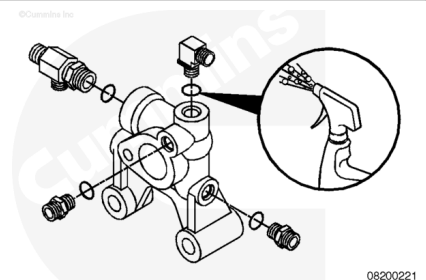
Install the turbocharger coolant return fitting.

Torque Value: 54 n•m [40 ft-lb]

Install the cylinder head vent fitting and the EGR valve coolant return fitting into the water manifold.

Torque Value: 16 n•m [140 in-lb]

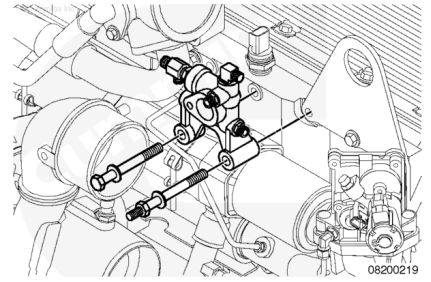
Install the EGR cooler coolant vent line T-fitting. Do **not** tighten the EGR cooler coolant vent line T-fitting until after the EGR cooler coolant vent lines have been installed and torqued to the T-fitting.



Install

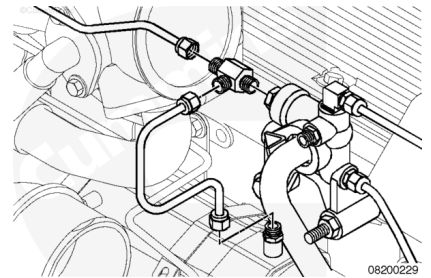
Start the mounting capscrew and loose washer and the studded capscrew and loose washer through the water manifold, lift bracket, and air intake in the rocker housing.

Do not torque the water manifold capscrews until all the EGR cooler coolant vent lines, the EGR valve return line, and the coolant vent lines have been installed.



Install the EGR cooler coolant vent lines. Refer to Procedure 011-031 (</qs3/pubsys2/xml/en/procedures/35/35-011-031-tr.html>).

Tighten the EGR cooler coolant vent line T-fitting to the water manifold.



Torque Value: 12 n•m [106 in-lb]

Install the coolant vent lines. Refer to Procedure 008-017 (</qs3/pubsys2/xml/en/procedures/35/35-008-017-tr.html>).

Install the EGR valve coolant return line. Refer to Procedure 011-030 (</qs3/pubsys2/xml/en/procedures/35/35-011-030-tr.html>).

Tighten the capscrew and studded capscrew of the water manifold.

Torque Value: 47 n•m [35 ft-lb]

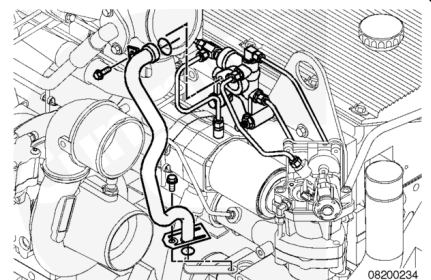
⚠ CAUTION ⚠

Do not use petroleum products on the coolant plumbing o-rings. The o-rings will swell and cause the connection to leak.

Install both o-rings into the water manifold return tube.

Install the water manifold return tube and capscrews to the thermostat housing support and the water manifold.

Torque Value: 47 n•m [35 ft-lb]



Finishing Steps

Install or connect the following:

- EGR cooler connection heat shield. Refer to Procedure 011-032 (</qs3/pubsys2/xml/en/procedures/35/35-011-032-tr.html>).
- Turbocharger coolant return line. Refer to Procedure 010-041 (</qs3/pubsys2/xml/en/procedures/35/35-010-041-tr.html>).
- Air inlet pipe to the turbocharger. Refer to Procedure 010-080 (</qs3/pubsys2/xml/en/procedures/35/35-010-080-tr.html>).
- Fill the cooling system. Refer to Procedure 008-018 (</qs3/pubsys2/xml/en/procedures/35/35-008-018-tr.html>).
- Operate the engine until it reaches a temperature of 82°C [180°F] and check for coolant leaks.

